

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023

EE1001-1 – BASIC ELECTRICAL ENGINEERING

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I

1. a) Define the terms (a) Phase (b) Average value (c) power factor (d) form factor (e) peak factor
- b) Using nodal analysis, find voltage across 20Ω resistor.

Marks BT* CO* PO*

5 L*1 1 1,2

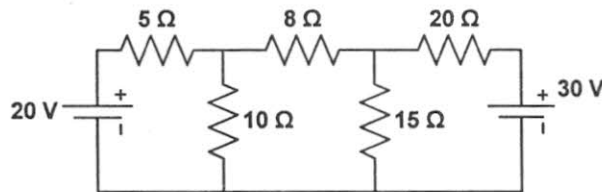


Figure 1b

5 L3 1 1,2

2. a) Define root mean square value of a sinusoidal alternating quantity. Derive an expression for RMS value of a sinusoidal alternating current.
- b) Using mesh analysis, determine the value of voltage across 8Ω resistor for the circuit shown in Figure 2b.

5 L2 1 1,2

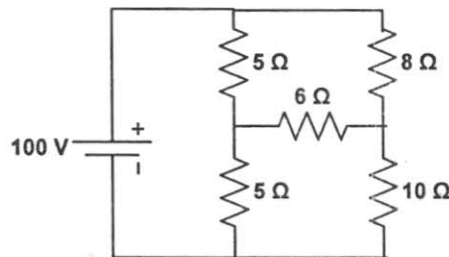


Figure 2b

5 L3 1 1,2

Unit – II

3. a) With neat circuit diagram and waveforms derive the relationship between voltage and current in a series RL circuit. Also derive the expression for the power consumed.
- b) With the help of a vector diagram show that two wattmeters are sufficient to measure the 3-phase power in a star connected load
4. a) Obtain the relationship between line and phase quantities of three phase delta connected load. Also derive the expression for three phase power.
- b) A pure resistance of 50Ω is in series with a capacitance of $100\mu\text{F}$. The series combination is connected across a 100V , 50Hz supply. Find impedance, current, power factor and power consumed by the circuit.

5 L2 2 1,2

5 L2 2 1,2

5 L2 2 1,2

5 L3 2 1,2

BT* Bloom's Taxonomy, L* Level ; CO* Course Outcome; PO* Program Outcome
